Problem statement & 3 steps of counter design :-

3-BIT SYNCHRONOUS COUNTER DESIGN

Step 1: Decide the no. & Type of Flip-Flops

Step 1: 3 Bit, FFs = 3, JK

Step 2: Excitation Table for the Flip-Flop

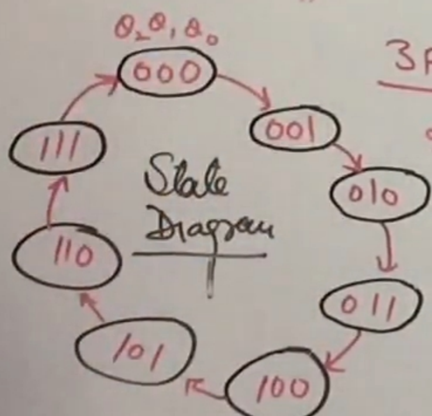
Step 3:- State Diagram & Ckt Exct. table

Step 4:- Obtain Boolean expression

Step 5:- Draw Ckt Diagram.

Step2

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0



3FF

$2^3 = 8$ States

Max Count = 8 - 1 = 7

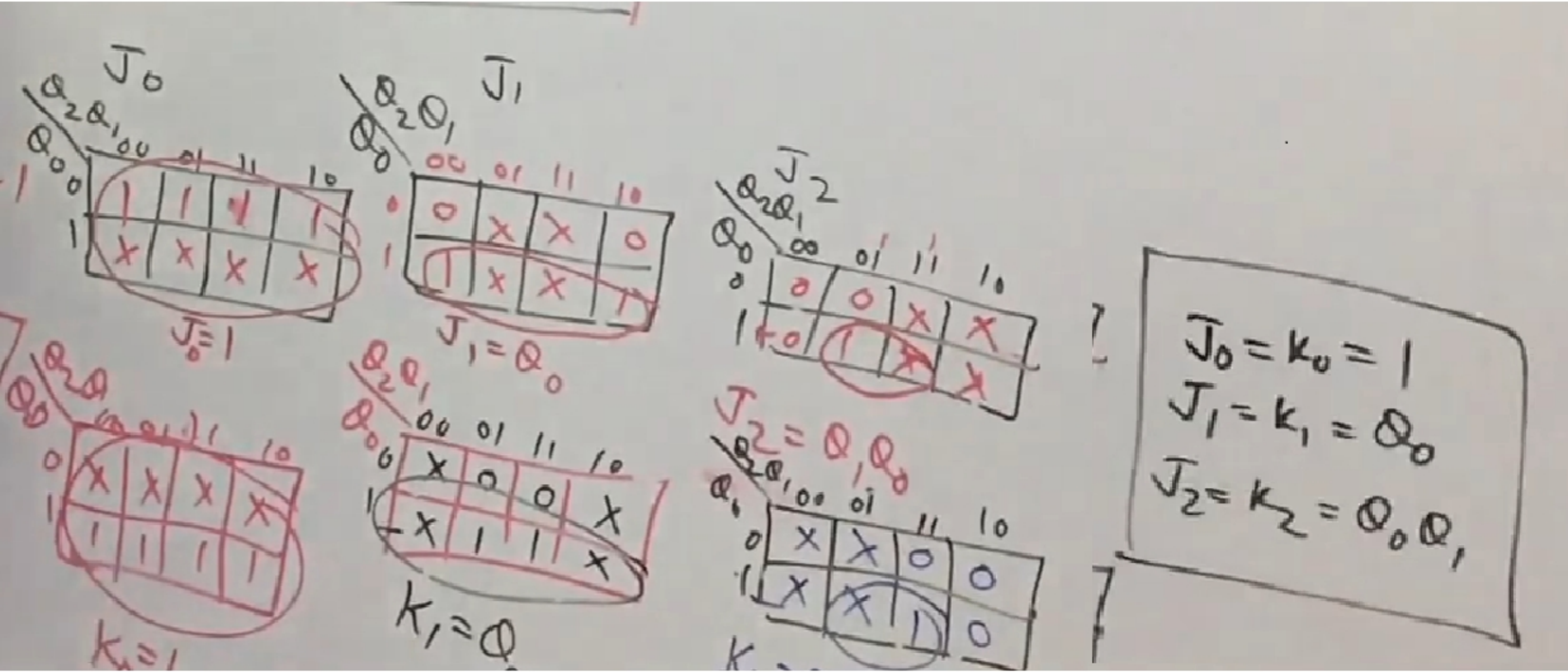
0 — 7

$\boxed{000 — 111}$

Excitation Table for ckt. :-

PS $Q_2 Q_1 Q_0$	N.S $Q_2 Q_1 Q_0$	$J_0 K_0$	$J_1 K_1$	$J_2 K_2$
0 0 0	0 0 1	1 X	0 X	0 X
0 0 1	0 1 0	X 1	1 X	0 X
0 1 0	0 1 1	1 X	X 0	0 X
0 1 1	1 0 0	X 1	X 1	1 X
1 0 0	1 0 1	1 X	0 X	X 0
1 0 1	1 1 0	X 1	1 X	X 0
1 1 0	1 1 1	1 X	X 0	X 0
1 1 1	0 0 0	X 1	X 1	X 1

K-map for circuit :-



Final ckt for 3-bit sync counter :-